

EmacEArt Stylized Grass Shader

User Guide -- 19th Century Courtyards

What is this shader?

The Stylized Grass Shader is a custom Unity shader included in this asset pack. It makes the grass on the terrain look alive -- it sways in the wind, shifts color gently over time, and responds to lighting in a natural, painterly way. It works with Unity's Universal Render Pipeline (URP) and requires no additional plugins.

Without this shader vs. with this shader

Without the shader	With the shader
Grass is flat and static -- no movement at all	Grass sways naturally in the wind
Single flat color, no visual depth	Dynamic color patches -- lighter and darker greens
Hard, sharp edges on each grass blade	Soft alpha-clipped edges, natural silhouette
Harsh shadows make the terrain look flat and muddy	Ambient lift softens shadows, keeps the scene warm and readable
No reaction to lighting or environment	Full PBR lighting -- reacts to sun, sky and shadows correctly

Features at a glance

Wind animation -- Grass blades bend and sway smoothly. You can control the direction, speed, strength, and frequency of the wind independently. A separate Gust layer adds irregular bursts of movement for a more natural, unpredictable feel.

Color variation -- Two independent tint layers (small-scale and large-scale) paint shifting patches of color across the grass. Each layer works in Blob mode (rounded patches) or Stripes mode (elongated streaks). Colors drift slowly over time, so the grass field never looks exactly the same twice.

Alpha cutout -- The texture's alpha channel defines the shape of each grass blade. Adjust the Alpha Cutoff slider to make blades thicker or more detailed.

Ambient lift -- A subtle but effective trick: it brightens the darkest areas of the grass slightly, preventing shadows from making the terrain look flat or muddy. You can tint these lifted areas to add warmth or coolness to the overall feel.

Normal map support -- An optional normal map adds fine surface detail that responds to light direction, giving the grass a more three-dimensional, textured appearance.

GPU Instancing -- Supports Unity GPU instancing -- thousands of individual grass meshes rendered efficiently. This keeps the frame rate high even on large outdoor terrain scenes.

How to adjust the shader

Select the material **EmacEArt_StylizedGrass_Mat** in the Project window. The Inspector shows all settings grouped by category. The default values are already tuned for this asset pack -- no changes are required to get the intended look. However, feel free to experiment:

- Lower **Wind Strength** for a calm, still day feel.
- Increase **Gust Strength** for a stormy or turbulent atmosphere.
- Change **Color Small / Color Large** to shift the overall color mood of the vegetation.
- Raise **Ambient Lift** if the grass looks too dark under your scene lighting.
- Adjust **Alpha Cutoff** if grass edges appear too jagged or too thick.